Propagator for Dirac field

For Klein-Gordon $(\partial_\mu \partial^\mu + m^2)\phi = 0$

$(\partial_\mu \partial^\mu + m^2)\, G(x) = \delta(x)$

$\frac{1}{p^2 - m^2 + i\varepsilon}$

$(i\gamma^\mu \partial_\mu - m)\psi = 0$ Dirac eq.

$(i\gamma^\mu \partial_\mu - m)\, G(x) = \delta(x)\, \mathbb{1}$ Green's function

Fourier transform $i\partial_\mu \to p_\mu$ $\qquad \hat{G}(p) = \int \frac{d^4x}{(2\pi)^4} e^{-ipx} G(x)$

$\delta(x) \to 1$

$\underbrace{(\gamma^\mu p_\mu - m\mathbb{1})}_{\text{matrix}} \hat{G}(p) = \mathbb{1}$

notation: $\gamma^\mu p_\mu = \not{p}$

$\not{\partial} = \gamma^\mu \partial_\mu$

$\not{A} = \gamma^\mu A_\mu$

$(\not{p} - m)\, \hat{G}(p) = 1$

$\frac{1}{2}(\gamma^\mu \gamma^\nu p_\mu p_\nu + \gamma^\nu \gamma^\mu p_\nu p_\mu)$

multiply by $\not{p} + m$

$\not{p}\not{p} = \gamma^\mu \gamma^\nu p_\mu p_\nu = \frac{1}{2}\underbrace{\{\gamma^\mu, \gamma^\nu\}}_{2\eta^{\mu\nu}\mathbb{1}} p_\mu p_\nu$

$= p_\mu p^\mu = p^2 = p_0^2 - \vec{p}^2$

$(\not{p} - m)(\not{p} + m) = \not{p}\not{p} - m^2$

$= p^2 - m^2$

$$\boxed{\hat{G}(p) = \frac{\not{p} + m}{p^2 - m^2 + i\varepsilon}}$$

denominator $\frac{1}{p_0^2 - (\vec{p}^2 +}$

$$G(x) = \lim_{\varepsilon \to 0^+} \int d^4p \, \frac{\not{p} + m}{p^2 - m^2 + i\varepsilon} e^{ip_\mu x^\mu}$$

Photon

$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$ $\qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \to F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$

EL eqs: $\partial_\mu F^{\mu\nu} = 0 \to \partial_\mu(\partial^\mu A^\nu - \partial^\nu A^\mu) = 0$

$\partial^2 = \partial_\mu \partial^\mu = \Box$ $\qquad \partial_\mu A^\mu = \eta^{\mu\nu} \partial_\mu A_\nu = \partial^\mu A_\mu$

$\partial^2 A^\nu - \partial^\nu \partial_\mu A^\mu = 0 \to \partial^2 A^\nu - \partial^\nu \partial^\mu A_\mu = 0$

$\to \partial^2 A_\mu \eta^{\mu\nu} - \partial^\nu \partial^\mu A_\mu = 0$

$$\boxed{= (\eta^{\mu\nu}\partial^2 - \partial^\mu \partial^\nu) A_\mu = 0}$$

$$(\eta^{\mu\nu}\partial^2 - \partial^\mu\partial^\nu)\, G_{\nu\lambda}(x) = \delta^\mu_\lambda\, \delta(x)$$ Green's function by definition

$K^{\mu\nu} = \eta^{\mu\nu}\partial^2 - \partial^\mu\partial^\nu$ matrix of diff. operators

$K^{\mu\nu} G_{\nu\lambda}(x) = \delta(x)\, \delta^\mu_\lambda$

$\delta_{\mu\nu} = \delta_{\mu\nu}$ $\quad \delta^{\mu\nu} = \delta^{\mu\nu}$

$\delta_{\mu\nu}\delta^{\nu\lambda} = \delta_\mu^\lambda$

We are going to impose Lorenz gauge

$\partial_\mu A^\mu = 0$ eq. becomes $\eta^{\mu\nu}\partial^2 A_\mu = 0$

$$\rightarrow \quad \eta^{\mu\nu}\partial^2 G_{\nu\lambda}(x) = \delta^\mu_\lambda\, \delta(x)$$

momentum space $\quad \partial^2 \rightarrow -p^2$

$-\eta^{\mu\nu} p^2 \hat{G}_{\nu\lambda}(p) = \delta^\mu_\lambda$ $\rightarrow$ multiply by $\eta_{\mu\sigma}$ on both sides

$$\boxed{\hat{G}_{\nu\lambda}(p) = -\frac{\eta_{\nu\lambda}}{p^2 + i\varepsilon}}$$

$$-\eta^{\mu\nu}\eta_{\mu\sigma}\, p^2 \hat{G}_{\nu\lambda}(p) = \delta^\mu_\lambda\, \eta_{\mu\sigma}$$

$$-\delta^\nu_\sigma\, p^2 \hat{G}_{\nu\lambda}(p) = \eta_{\lambda\sigma}$$

$$-p^2 \hat{G}_{\sigma\lambda}(p) = \eta_{\lambda\sigma}$$